Final Exam

● Graded

Student

Boyuan Huang

Total Points

43.5 / 45 pts

Question 1

Overall Grades **43.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 1.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Boyuan Huang.	hh304.	黄伯远

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 黄伯远

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	4	5	5	5	5	43.5

Problem 1. [5'] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

$\frac{9}{2} - 4$.

~~26~~

~~26~~

$22 - 21 = 1$

$\begin{cases} -14 + 15 = 1. \\ 11 - 10 = 1. \end{cases}$

$$\begin{bmatrix} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{bmatrix}.$$

$R_1 \leftrightarrow R_3$

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{bmatrix}.$$

$R_3 \leftarrow R_3 - 2R_1$

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & -3 & -4 & -1 \end{bmatrix}.$$

$R_2 \leftarrow R_2 - \frac{3}{2}R_2$

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 0 & \frac{1}{2} & -\frac{5}{2} \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 3 & -1 \\ 0 & 0 & 1 & -5 \end{bmatrix}.$$

$\therefore z = -5, 2y = -1 - 3z = 14 \Rightarrow y = 7, x = 1 - 2z = 11$

$(x, y, z) = (11, 7, -5)$.

4.5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1). $A \sim \begin{bmatrix} 2 & 4 & 2 \\ 0 & 10 & 10 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 2 & 1 \\ 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Columns 1 and 2 are pivot, hence a basis of col(A) is $\{[2\ -5\ 0\ 1]^T, [4\ 0\ 1\ 3]^T\}$.

(2.) From the ref form of A in (1), a basis of row(A) is $\{[1\ 2\ 1]^T, [0\ 1\ 1]^T\}$. 0.5

(3.) Let $Ax = 0$. where $x = [x_1\ x_2\ x_3]^T$

From (1), $\text{ref}(A) = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, let $x_3 = m$,

$x_2 = -m$, $x_1 = -2x_2 - x_3 = 2m - m = m$.

$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} m \\ -m \\ m \end{bmatrix} = m \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} (m \in R)$.

Hence, a basis of ker(A) is $\{[-1\ -1\ 1]\}$.

5/5

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$$V = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ -1 & -1 & -1\end{bmatrix}, \quad [V|I_3] = \left[\begin{array}{ccc|ccc}0 & 1 & 0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0 & 1 & 0\\ -1 & -1 & -1 & 0 & 0 & 1\end{array}\right].$$

$$[V|I_3] \sim \left[\begin{array}{ccc|ccc}1 & 0 & 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 1 & 0 & 0\\ -1 & -1 & -1 & 0 & 0 & 1\end{array}\right] \sim \left[\begin{array}{ccc|ccc}1 & 0 & 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 1 & 0 & 0\\ 0 & 0 & -1 & 1 & 1 & 1\end{array}\right].$$

$$\sim \left[\begin{array}{ccc|ccc}1 & 0 & 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 1 & 0 & 0\\ 0 & 0 & 1 & -1 & -1 & -1\end{array}\right] = [I_3|A^{-1}].$$

Let $W = \begin{bmatrix}1 & 0 & 2\\ 1 & 2 & 2\end{bmatrix}$, we have $A.V = W \Rightarrow A = WV^{-1}$.

$$\therefore A = \begin{bmatrix}1 & 0 & 2\\ 1 & 2 & 2\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ -1 & -1 & -1\end{bmatrix} = \begin{bmatrix}-2 & -1 & -2\\ 0 & -1 & -2\end{bmatrix}.$$

Problem 4. [5'] Find the Jordan canonical form of

$$A=\begin{bmatrix}1&2&0\\0&2&0\\-2&-2&1\end{bmatrix},\quad A-\lambda I=\begin{bmatrix}1-\lambda&2&0\\0&2-\lambda&0\\-2&-2&1-\lambda\end{bmatrix}$$

$$P_A(\lambda)=\det(A-\lambda I).$$
$$=(-1)^4(2-\lambda)\det\left(\begin{bmatrix}1-\lambda&0\\-2&1-\lambda\end{bmatrix}\right)$$
$$=(2-\lambda)(1-\lambda)^2.$$

$\lambda=2,1,1.$

① $\lambda=1$ $(AM=2)$.

$$A-I=\begin{bmatrix}0&2&0\\0&1&0\\-2&-2&0\end{bmatrix}\sim\begin{bmatrix}1&1&0\\0&1&0\\0&0&0\end{bmatrix}.\quad (GM=1).$$

let $x_3=m$, $x_2=0$, $x_1=-x_2=0$. $\therefore$ an eigenvector $V_1=\begin{bmatrix}0\\0\\1\end{bmatrix}$.

Let ~~V2 be~~ V_2 be a generalized eigenvector with $\lambda=1$,

$$(A-I)V_2=V_1.$$

$$\left[\begin{array}{ccc|c}0&2&0&0\\0&1&0&0\\-2&-2&0&1\end{array}\right]\sim\left[\begin{array}{ccc|c}1&1&0&-\frac{1}{2}\\0&1&0&0\\0&0&0&0\end{array}\right],\quad \begin{cases}x_1=-\frac{1}{2}-x_2=-\frac{1}{2}\\x_2=0\\x_3=n\end{cases}$$

set $n=1\Rightarrow V_2=\begin{bmatrix}-\frac{1}{2}\\0\\1\end{bmatrix}$, $J_2(1)=\begin{bmatrix}1&1\\0&1\end{bmatrix}$.

② $\lambda=2\,(AM=1)$.

$$A-2I=\begin{bmatrix}-1&2&0\\0&0&0\\-2&-2&-1\end{bmatrix}\sim\begin{bmatrix}2&2&1\\-1&2&0\\0&0&0\end{bmatrix}\sim\begin{bmatrix}1&0&\frac{1}{6}\\-1&2&0\\0&0&0\end{bmatrix}\sim\begin{bmatrix}1&-2&0\\0&6&1\\0&0&0\end{bmatrix},$$

$x_3=q\Rightarrow x_2=\frac{-q}{6}$, $x_1=2x_2=\frac{-q}{3}$, set $q=6$, $V_3=\begin{bmatrix}-2\\-1\\6\end{bmatrix}$.

$J_1(2)=[2]$.

$$\therefore J=\operatorname{diag}(J_2(1),J_1(2))=\begin{bmatrix}1&1&0\\0&1&0\\0&0&2\end{bmatrix},\quad P=\begin{bmatrix}0&-\frac{1}{2}&-2\\0&0&-1\\1&1&6\end{bmatrix}.$$

$$A=PJP^{-1}$$

4/5

$\lambda^2-7\lambda+10$ $\quad \begin{matrix} 1 & -2 \\ 1 & -5 \end{matrix}$.

oxy.

Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2+3y^2+4z^2+2xz-2yz. \qquad v=\begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

$f(v) = v^T A v.$

$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$, $A-\lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$.

$P_A(\lambda) = (-1)^4(3-\lambda)\det\left(\begin{bmatrix} 3-\lambda & 1 \\ 1 & 4-\lambda \end{bmatrix}\right) + (-1)^5(-1)\det\left(\begin{bmatrix} 3-\lambda & 0 \\ 1 & -1 \end{bmatrix}\right)$.

$= (3-\lambda)\left[(3-\lambda)(4-\lambda)-1\right] + \lambda - 3.$

$= (3-\lambda)\left[\lambda^2-7\lambda+12-2\right)$

$= (3-\lambda)(\lambda-2)(\lambda-5).$ $\quad \lambda = 3, 2, 5.$

$\lambda=2$: $A-2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$, $\begin{cases} x_3 = m, \\ x_2 = m \\ x_1 = -m. \end{cases}$

set $m=1$

$V_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$.

$\lambda=3$: $A-3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $\begin{cases} x_3 = 0. \\ x_2 = m. \\ x_1 = m \end{cases}$, set $m=1$.

$V_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

$\lambda=5$: $A-5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 0 & -2 & -1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. $\begin{cases} x_3 = m \\ x_2 = -\frac{m}{2} \\ x_1 = \frac{m}{2}, \end{cases}$

set $m=2$, $V_3 = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$.

$\therefore D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$, $\begin{bmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix}$, $A = PDP^{-1}$

$P = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix}$. -1

~~$(1-1)(1-\frac{-1}{2})(\frac{2}{2})$~~.

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

Let $V = [v_1\ v_2\ v_3] = \begin{bmatrix} 0 & 2 & -1 \\ 2 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix}$, $x = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$,

$Vx = w$.

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \sim \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 1 & 1 & 0 \end{array}\right].$$

$$\sim \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & \frac{3}{2} & \frac{3}{2} \end{array}\right]$$

~~$\therefore (a,b,c) = (\frac{5}{2}, -2, 1)$~~.

$$\begin{cases} c = 1 \\ b = \frac{c-3}{2} = -1 \\ a = \frac{8-c+b}{2} = 3. \end{cases}$$

$\therefore (a,b,c) = (3,-1,1)$

3/5

$\frac{\sqrt{2}}{2} = \frac{5}{\sqrt{2}}$.

$\frac{10+36+4}{4} = \frac{50.25}{4\times 2}$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

Apply Gram-S. with $B = \left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}$,

$u_1 = b_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$,

$u_2 = b_2 - \frac{b_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{1+1}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$.

$= \begin{bmatrix}3\\-1\\-1\end{bmatrix}$ $= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$= [3 \quad -1 \quad -1]^T$

Normalize: $q_1 = \frac{u_1}{\|u_1\|} = \frac{u_1}{\sqrt{2}} = [0 \quad \frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \quad 0]^T$

~~$q_2 = \frac{u_2}{\|u_2\|} = \frac{u_2}{\sqrt{9+\frac{1}{4}+\frac{9}{4}+1}} = \frac{u_2}{\frac{\sqrt{12}}{2}} = \frac{\sqrt{12}}{5}$~~

$= [\frac{3\sqrt{2}}{5} \quad -\frac{\sqrt{2}}{10} \quad \frac{3\sqrt{2}}{10} \quad -\frac{\sqrt{2}}{5}]^T$

$q_2 = \frac{u_2}{\|u_2\|} = \frac{u_2}{\sqrt{9+1+1+1}} = \frac{1}{2\sqrt{3}} u_2 = [\frac{3}{2\sqrt{3}} \quad -\frac{1}{2\sqrt{3}} \quad \frac{1}{2\sqrt{3}} \quad -\frac{1}{2\sqrt{3}}]^T$

∴ An orthonormal basis for V is.

$B = \{q_1, q_2\} = \{[0 \quad \frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \quad 0]^T, [\frac{3}{2\sqrt{3}} \quad -\frac{1}{2\sqrt{3}} \quad \frac{1}{2\sqrt{3}} \quad -\frac{1}{2\sqrt{3}}]^T\}$.

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

$$A = -A^T.$$

$$\begin{aligned}
&\det(A) \\
&= \det(-A^T) \\
&= \det((-1)A^T) \\
&= (-1)^n \det(A^T) \\
&= (-1)^n \det(A)
\end{aligned}$$

If n is odd, $(-1)^n = -1$,

$\therefore \det(A) = -\det(A) \Rightarrow 2\det(A) = 0 \Rightarrow \det(A) = 0.$

Q.E.D.

5/5

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

Since $\|\vec{u}+\vec{v}\|, \|\vec{u}-\vec{v}\| \geq 0$,

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$$

$$\Downarrow$$

$$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2.$$

$$\vec{u}^2 + 2\vec{u}\cdot\vec{v} + \vec{v}^2 = \vec{u}^2 - 2\vec{u}\cdot\vec{v} + \vec{v}^2.$$

$$4\vec{u}\cdot\vec{v} = 0.$$

$$\vec{u}\cdot\vec{v} = 0. \Rightarrow \vec{u} \perp \vec{v}.$$

Q.E.D.